

Lecture 6 – Data Observation Techniques

INT104 Artificial Intelligence

Data Observation Techniques



PCA, Correlation and Regression

Dr. Shengchen Li

Associate Professor

Department of Intelligence Science

Xi'an Jiaotong–Liverpool University

Too many features

- **Dimensionality Reduction:**

Reduce complex datasets to fewer variables while preserving critical information.

- **Visualization:**

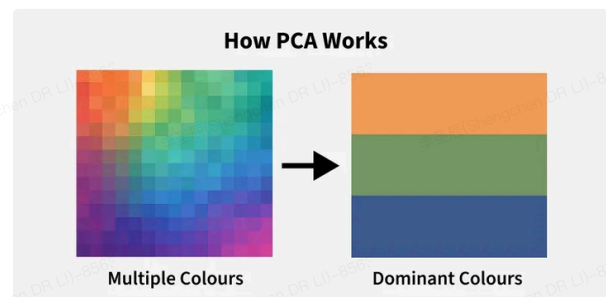
Project high-dimensional data onto 2D/3D plots for intuitive exploration.

- **Noise Reduction:**

Discard low-variance components (often noise).

- **Multicollinearity Mitigation:**

Create orthogonal (uncorrelated, independent) features for downstream modeling.

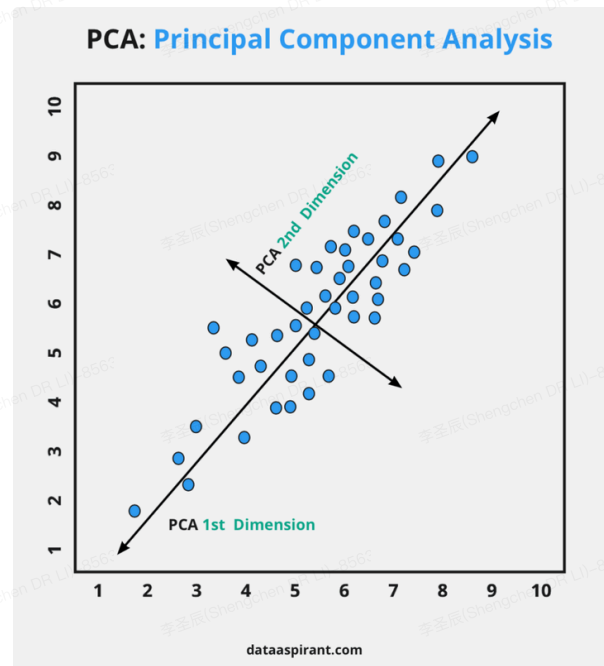


How PCA Works

The Core Idea

PCA finds new axes (**principal components**) that:

1. Maximize variance in the data.
2. Are mutually orthogonal (uncorrelated).



PCA Simplified

1. Standardise Data:

Scale features (mean=0, std=1).

2. Compute Covariance Matrix:

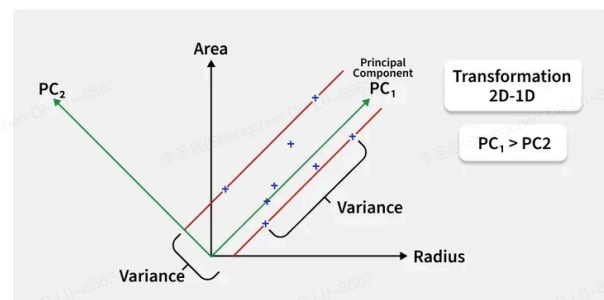
Quantify feature relationships.

3. Eigen Decomposition:

Extract eigenvectors (PC directions) and eigenvalues (variance explained).

4. Project Data:

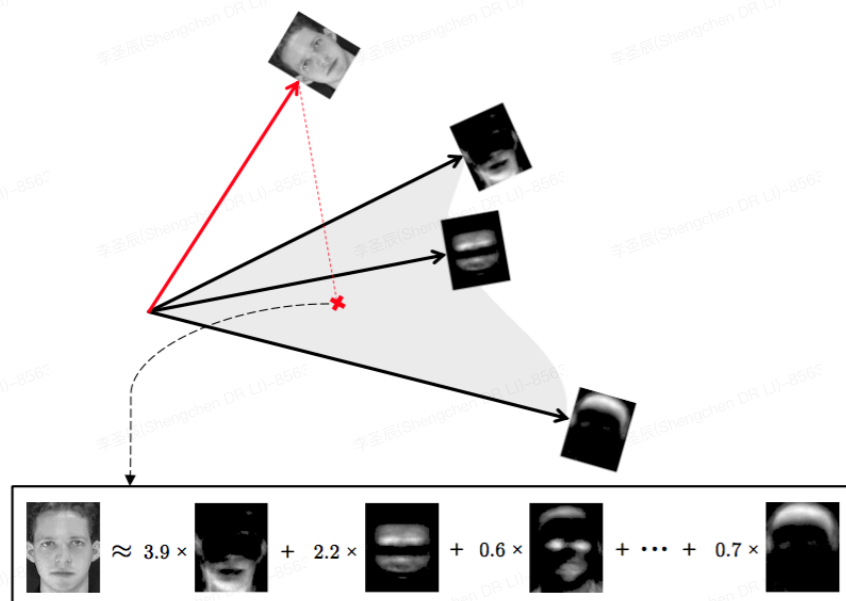
Transform data onto selected PCs.



PCA

$$X = W^T H$$

H is a basis



PCA

- **Linearity Assumption:**

PCA captures linear relationships only.

- **Scale Sensitivity:**

Standardisation is critical (use correlation matrix for unit-varient features).

- **Interpretability:**

PCs are mathematical constructs; domain context is needed to assign meaning.

- **Outlier Vulnerability:**

Extreme values skew variance maximization.

deepnote.com

<https://deepnote.com/app/auditory-intelligence-computing-0b7e/INT104-2025-530324ac-020c-456b-8be5-a3faffa0d0e...>

Real-World Applications

Where PCA Shines

- **Genomics:**

Reduce 20,000 genes to key components driving disease.

- **Finance:**

Identify latent market risk factors from asset returns.

- **Computer Vision:**

Eigenfaces for facial recognition (compressing pixel data).

- **Anomaly Detection:**

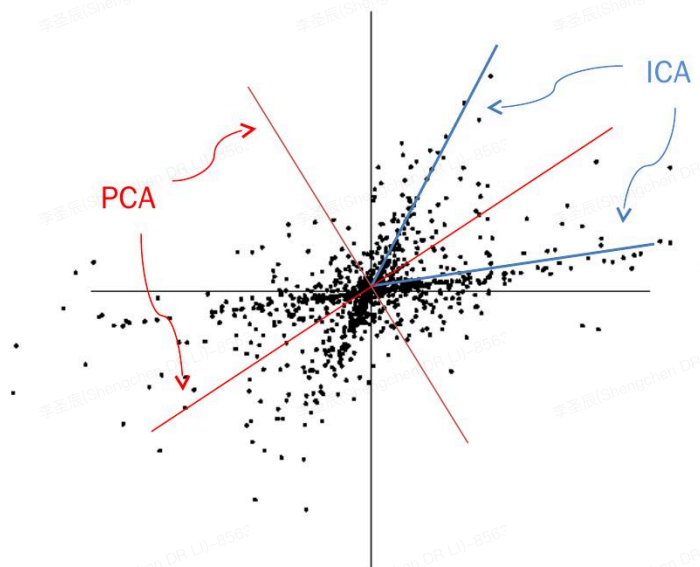
Outliers deviate from dominant PCs.

Case Study:

How do you determine the value of k in GMM?

Independent Component Analysis

PCA vs ICA

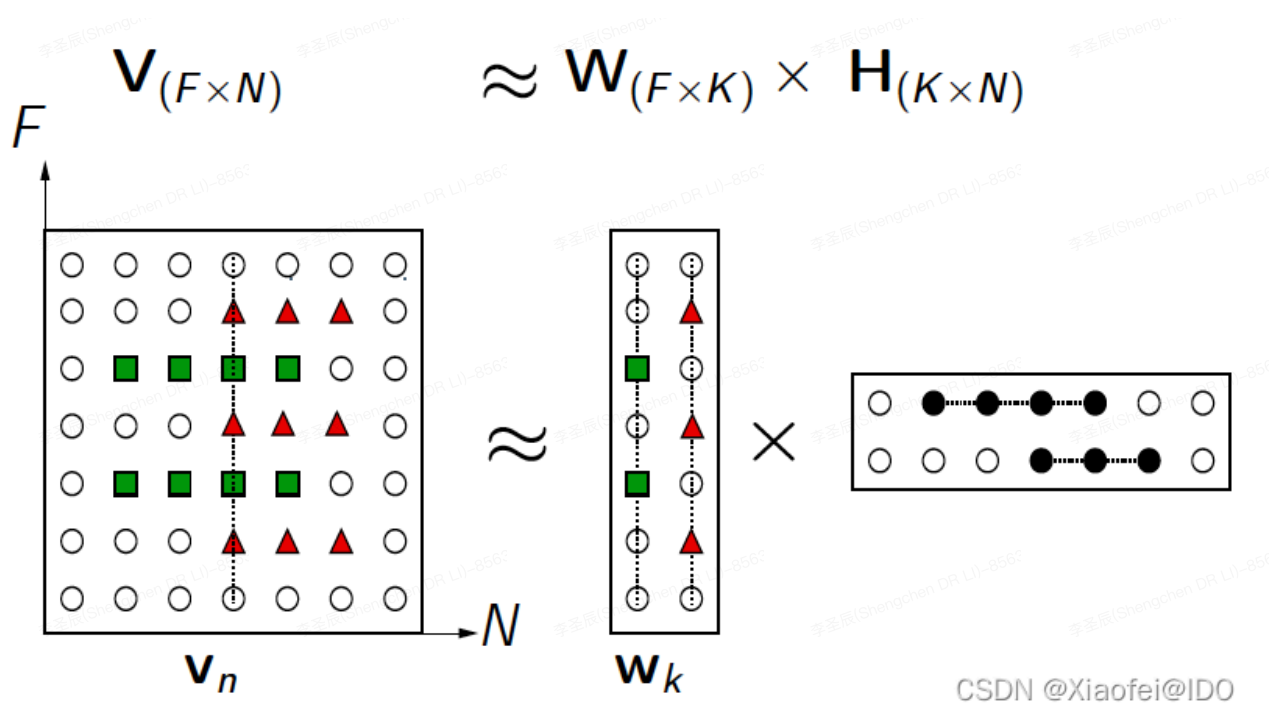


University of Patras
Computer Vision
Group



ELLAB
UNIVERSITY OF PATRAS

Non-negative Matrix Factorisation



Correlation

- What is the correlation?

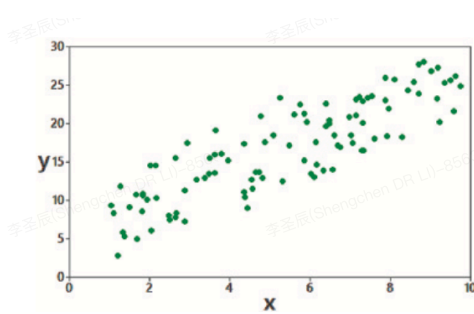


A correlation exists between two variables when the values of one variable are somehow associated with the values of the other variable.

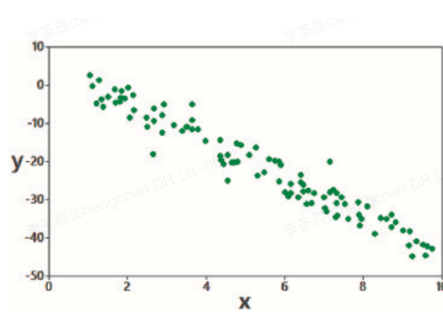


A linear correlation exists between two variables when there is a correlation and the plotted points of paired data result in a pattern that can be approximated by a straight line.

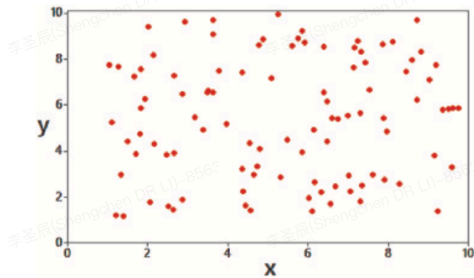
Identify Correlation



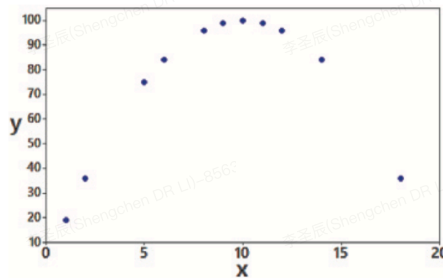
(a) Positive correlation: $r = 0.859$



(b) Negative correlation: $r = -0.971$



(c) No correlation: $r = 0.074$



(d) Nonlinear relationship: $r = 0.330$

Pearson's Correlation Coefficient

Example of Correlation Coefficient

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{n(\sum x^2 - (\sum x)^2)} \sqrt{n(\sum y^2 - (\sum y)^2)}}$$

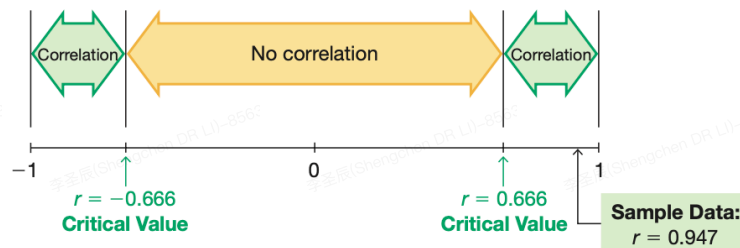


FIGURE 10-3 Critical r Values and the Computed r Value

Properties of the Linear Correlation Coefficient

1. The value of r is always between -1 and 1 inclusive. That is, $-1 \leq r \leq 1$.
2. If all values of either variable are converted to a different scale, the value of r does not change.
3. The value of r is not affected by the choice of x or y . Interchange all x values

and y values, and the value of r will not change.

4. r measures the strength of a linear relationship. It is not designed to measure the strength of a relationship that is not linear.
5. r is very sensitive to outliers in the sense that a single outlier could dramatically affect its value.

Spurious Correlation

- Does high correlation value leads to causality?
- Does averaged correlation coefficients lead to the (non-)existence of correlation?
- Does a low correlation coefficient lead to the lack of correlation?



There are other types of correlation coefficients

Regression

Given a collection of paired sample data, the regression line (or line of best fit, or least-squares line) is the straight line that “best” fits the scatterplot of the data.



Now think about how to measure the "fitness" in the definition

$$\hat{y} = b_0 + b_1 y$$

Regression

Given

$$\hat{y} = b_0 + b_1 y$$

we have

$$b_1 = r \frac{s_y}{s_x}$$

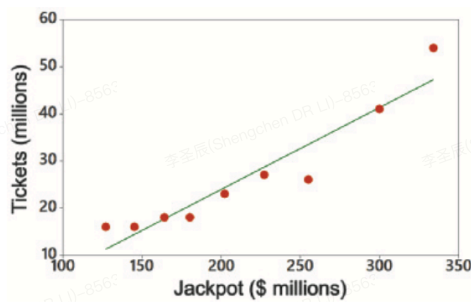
$$b_0 = \bar{y} - b_1 \bar{x}$$

Outliers and Influential Points

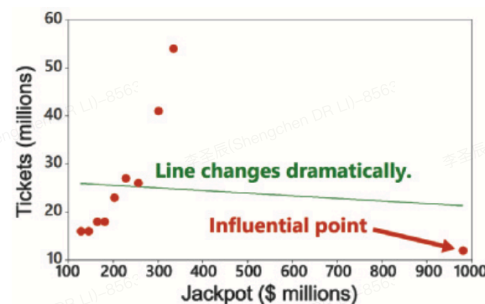
 In a scatterplot, an outlier is a point lying far away from the other data points.

 Paired sample data may include one or more influential points, which are points that strongly affect the graph of the regression line.

Original Jackpot/Ticket Data from Table 10-1



Jackpot/Ticket Data with Additional Point: (980, 12)



- How do we find out outliers?

Residuals

 For a pair of sample x and y values, the residual is the difference between the observed sample value of y and the y value that is predicted by using the regression equation. That is,

$$\text{Residual} = \text{Observed } y - \text{Predicted } y = y - \hat{y}$$

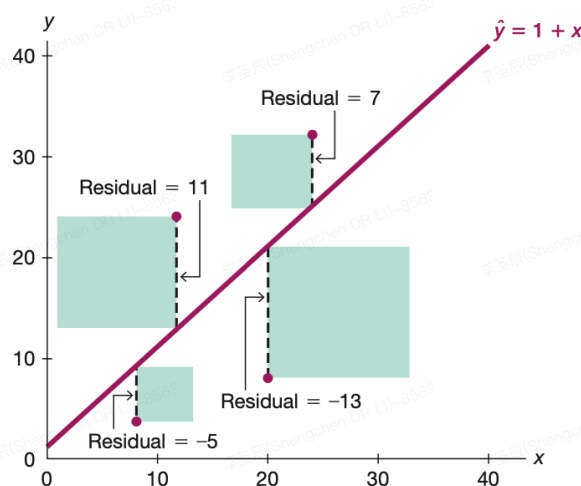


FIGURE 10-6 Residuals and Squares of Residuals



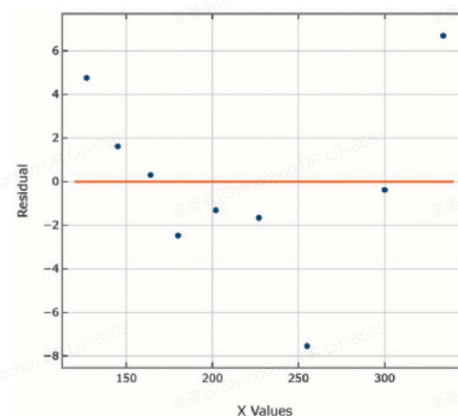
A straight line satisfies the least-squares property if the sum of the squares of the residuals is the smallest sum possible.

Residual Plot



A residual plot is a scatterplot of the (x, y) values after each of the y -coordinate values has been replaced by the residual value $y - \hat{y}$ (where $\hat{y}$ denotes the predicted value of y). That is, a residual plot is a graph of the points $(x, y - \hat{y})$.

Statdisk



What residual plot do we pursue?

Multiple Regression Equation

A multiple regression equation expresses a linear relationship between a response variable y and two or more predictor variables $(x_1, x_2, \dots, x_k)$. The general form of a multiple regression equation obtained from sample data is

$$\hat{y} = b_0 + b_1x + b_2x + \dots + b_kx_k$$

Logistic Regression

- Logistic Regression uses the **Logistic Function** to transform a linear combination of input features into a probability between 0 and 1.

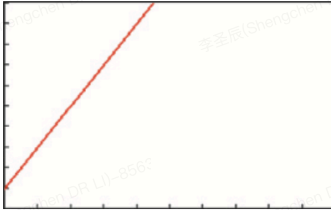
$$P(Y = 1) = \frac{1}{(1 + e^{-z})}$$

Where $z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p$ (the linear combination)

Non-linear Regression

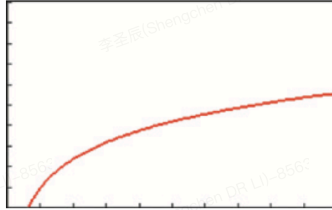
Linear: $y = a + bx$

Example: $y = 1 + 2x$



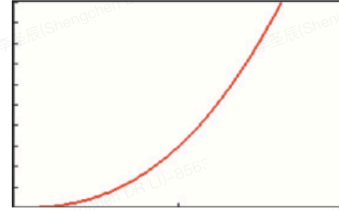
Logarithmic: $y = a + b \ln x$

Example: $y = 1 + 2 \ln x$



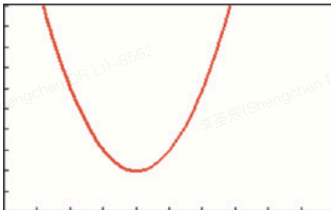
Power: $y = ax^b$

Example: $y = 3x^{2.5}$



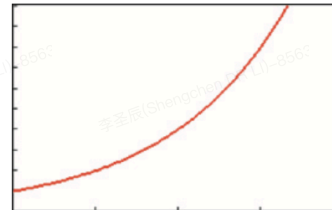
Quadratic: $y = ax^2 + bx + c$

Example: $y = x^2 - 8x + 18$



Exponential: $y = ab^x$

Example: $y = 2^x$



Discussion



In which case we could use regression?

- Can we use regression for classification?
- What if we need classification with multiple-class?